THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET A

Chapter 7: **Coordinate Geometry** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 10	$\sqrt{36+64}=10$
2	**(a)** $4\sqrt{2}$	$\sqrt{16+16}=4\sqrt{2}$
3	**(a)** $(4,1)$	$\left(\frac{2+6}{2},\frac{-3+5}{2}\right)$
4	**(b)** $(a,0)$	Its ordinate is 0
5	**(b)** Incorrect — it lies on the y-axis	Abscissa 0 ⇒ on y-axis
6	**(a)** 1 : 2	$\frac{6k-3}{k+1}=0\Rightarrow k=\frac{1}{2}$
7	**(a)** Student A is correct	$\sqrt{9+16}=5$; B added $3+4$
8	**(a)** −12	Mid-point $(-4,4)$ ⇒ $\frac{a}{3}=-4$
9	**(a)** $(3,5)$	$\left(\frac{8+1}{3},\frac{12+3}{3}\right)$
10	**(a)** ±4	$9+p^2=25$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\sqrt{(2a)^2+(2b)^2}=2\sqrt{a^2+b^2}$	1+1
12	$(0,y)$: $36+(5-y)^2=16+(3-y)^2\Rightarrow 61-10y=25-6y\Rightarrow y=9$ ⇒ $(0,9)$	1+1
13	$\left(\frac{3\times8+4}{4},\frac{3\times5-3}{4}\right)=(7,3)$	1+1
14	$AB^2=25,\ AC^2=25,\ BC^2=50$ ⇒ $AB=AC$ and $AB^2+AC^2=BC^2$ ⇒ claim is **correct**	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\frac{6k-3}{k+1}=-1\Rightarrow k=\frac{2}{7}$ ⇒ 2 : 7. (Check y: $\frac{-16+70}{9}=6$ ✓)	1+1+1
16	$(x-3)^2+(y-6)^2=(x+3)^2+(y-4)^2\Rightarrow -12x-4y+20=0\Rightarrow 3x+y-5=0$	1+1+1
17	Mid-point $(0,5)$; quarter points $\left(-1,\frac{7}{2}\right),(0,5),\left(1,\frac{13}{2}\right)$	1+1+1
	OR: Diagonals bisect each other: $\left(\frac{1+x}{2},4\right)=\left(\frac{7}{2},\frac{5+y}{2}\right)$ ⇒ $x=6,\ y=3$	1+1+1
18	$AB^2=BC^2=CD^2=DA^2=34$ and diagonals $AC^2=BD^2=68$ ⇒ equal sides, equal diagonals ⇒ square	2+1
	OR: $\frac{-k+5}{k+1}=0\Rightarrow k=5$ ⇒ 5 : 1; $y=\frac{-20-6}{6}=-\frac{13}{3}$ ⇒ $\left(0,-\frac{13}{3}\right)$	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	P (1 : 2) $=\left(2,-\frac{5}{3}\right)$, Q (2 : 1) $=\left(0,-\frac{7}{3}\right)$ (3). $AP=\sqrt{4+\frac{4}{9}}=\frac{2\sqrt{10}}{3}=PQ$ ✓ (2)	3+2
	OR: (i) Each side $=\sqrt{18}=3\sqrt{2}$ (ii) $AC=BD=6$ (iii) Equal sides and equal diagonals ⇒ square; **Champa** is correct	2+2+1
20	(i) $\sqrt{64+36}=10$ units (ii) $P=\left(\frac{10+6}{4},\frac{9+9}{4}\right)=(4,4.5)$ (iii) $M=(6,6)$ (iv) $MS=MH=\sqrt{16+9}=5$ ✓	1+2+1+1
	OR: (i) $A=(2\cdot2-1,\ 2(-3)-4)=(3,-10)$ (ii) $r=\sqrt{1+49}=5\sqrt{2}$ (iii) distance from centre $=7\neq5\sqrt{2}$ ⇒ does not lie on it	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 7: **Coordinate Geometry** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 13	$\sqrt{25+144}=13$
2	**(a)** 39	$\sqrt{1296+225}=39$
3	**(a)** $(0,-3)$	$\left(0,\frac{-6}{2}\right)$
4	**(b)** on the negative x-axis	Ordinate 0, abscissa negative
5	**(b)** Incorrect — the mid-point divides it in the ratio 1 : 1	Mid-point ⇒ equal parts
6	**(a)** 2 : 1	$\frac{3k}{k+1}=2\Rightarrow k=2$
7	**(a)** Student A is correct	$\sqrt{16+9}=5$
8	**(a)** $(2,6)$	$x+6=8,\ 4+y=10$
9	**(a)** $(5,4)$	$D=A+C-B=(5,4)$
10	**(a)** 12	$3+4+5=12$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$(x-3)^2+9=25\Rightarrow x-3=\pm4\Rightarrow x=7$ or -1	1+1
12	$\left(\frac{12-2}{5},\frac{-9+14}{5}\right)=(2,1)$	1+1
13	$(x-2)^2+25=(x+2)^2+81\Rightarrow -8x=56\Rightarrow x=-7$ ⇒ $(-7,0)$	1+1
14	$AB=\sqrt{5},\ BC=\sqrt{212},\ AC=\sqrt{265}$; $\sqrt{5}+\sqrt{212}\approx16.8\neq16.3$. Not collinear — claim is **false**.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\frac{2k+\frac{1}{7}}{k+1}=\frac{3}{4}\Rightarrow 8k+2=3k+3\Rightarrow k=\frac{1}{5}$ ⇒ 1 : 5	1+1+1
16	$(x-7)^2+(y-1)^2=(x-3)^2+(y-5)^2\Rightarrow -8x+8y+16=0\Rightarrow x-y=2$	1+1+1
17	$AB=BC=CD=DA=2\sqrt{2},\ AC=BD=4$ ⇒ **square**	2+1
	OR: $AB=8,\ AC=\sqrt{16+16}=4\sqrt{2},\ BC=4\sqrt{2}$ ⇒ isosceles; $AC^2+BC^2=64=AB^2$ ⇒ right-angled at C; perimeter $=8+8\sqrt{2}$ units	2+1
18	$p=\frac{-5+11}{2}=3$; $AB=\sqrt{64+256}=\sqrt{320}=8\sqrt{5}$	1+2
	OR: $(1:2)\ \left(\frac{-7+4}{3},\frac{4-4}{3}\right)=(-1,0)$; $(2:1)\ \left(\frac{-14+2}{3},\frac{8-2}{3}\right)=(-4,2)$	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$P=\left(\frac{24+6}{5},\frac{14-9}{5}\right)=(6,1)$ (2½). New A′(6, −3), B′(12, 2): $P'=\left(\frac{24+18}{5},\frac{4-9}{5}\right)=\left(\frac{42}{5},-1\right)$ (2½)	2½+2½
	OR: $\frac{5k-5}{k+1}=0\Rightarrow k=1$ ⇒ 1 : 1 (2). Point $=\left(\frac{-4+1}{2},0\right)=\left(-\frac{3}{2},0\right)$ (2); it is the mid-point ✓ (1)	2+2+1
20	(i) Green $(2,25)$, red $(8,20)$ (ii) $\sqrt{36+25}=\sqrt{61}$ m (iii) mid-point $(5,22.5)$: on the 5th line, 22.5 m along	2+2+1
	OR: (i) $D=\left(\frac{5}{2},\frac{11}{2}\right)$, $E=\left(\frac{11}{2},4\right)$ (ii) $DE=\sqrt{9+\frac{9}{4}}=\frac{3\sqrt{5}}{2}$, $BC=\sqrt{36+9}=3\sqrt{5}$ ✓ (iii) Mid-point theorem	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.